Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which three broad needs define habitability for life as we know it?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Oxygen, sunlight, and soil

B. A magnetic field, rings, and moons

C. Ice, methane, and nitrogen only

D. Liquid water, energy, and key chemical elements

Correct: D

2. (2 points) A planet moves from 1 AU to 2 AU. By what factor does the starlight flux change?

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

A. It becomes 1/8 as large

B. It becomes 1/4 as large

C. It becomes 1/2 as large

D. It doubles

Correct: B

3. (2 points) A body orbits the Sun at 9 AU. What is its period?

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

A. 9 years

B. 81 years

C. 3 years

D. 27 years

Correct: D

4. (2 points) Why is Venus's surface hotter than Mercury's?

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

A. Venus has active tidal heating

B. A thick CO_2 atmosphere creates a runaway greenhouse

C. Venus is closer to the Sun

D. Venus has a strong magnetic field

Correct: B

5. (2 points) Which moon has its own internally generated magnetic field?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Europa

B. Titan

C. Enceladus

D. Ganymede

Correct: D

6. (2 points) Which mission returned a sample of asteroid Bennu?

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. Kepler

B. OSIRIS-REx

C. Rosetta

D. Dragonfly

Correct: B

7. (2 points) Which method directly gives a planet's radius?

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
Star: An ultracool red dwarf, about 40 light-years away.
Sizes: All planets are roughly Earth-sized.
JWST: The two innermost planets show no thick atmosphere.

A. Radial velocity

B. Microlensing alone

C. Astrometry

D. Transit

Correct: D

8. (2 points) Why are small, hot worlds poor at keeping light gases?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Gravity is too strong

B. Light gases freeze out

C. The Sun adds gas

D. Gas molecules move fast compared with a low escape velocity

Correct: D

9. (2 points) If a star’s surface temperature doubles, its emitted power per area changes by:

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth’s 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

A. 8 times

B. 16 times

C. 2 times

D. 4 times

Correct: B

10. (2 points) If the distance between two masses triples, the gravitational force becomes:

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth’s mass and 2 times Earth’s radius. Compare its surface gravity and escape velocity with Earth’s.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

A. 1/3 as large

B. 3 times as large

C. 1/27 as large

D. 1/9 as large

Correct: D

11. (2 points) In a simplified habitable-zone model, stellar luminosity increases by a factor of four. By what factor must distance increase to preserve incident flux?

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , .

12. (2 points) A synthetic planet and star have radius ratio 1/100. For a uniform stellar disk and central transit, what fraction of light is blocked?

answer: 0.0001 (2 points).

Numerical target 0.0001; absolute tolerance 2e-07. Accepted units: , .

13. (2 points) Compare surface-water and subsurface-water habitability arguments using Europa or Enceladus as listed examples.

answer: Surface conditions may be hostile while subsurface liquid can be maintained by internal/tidal heating. Liquid, chemistry and usable energy must all be considered; water alone does not prove life. (2 points).

14. (2 points) Titan has surface liquid hydrocarbons and water ice. Explain why habitability questions should specify the solvent and biological assumptions.

answer: Habitability depends on the proposed chemistry and solvent; the presence of one kind of liquid does not automatically imply conditions suitable for familiar water-based life. (2 points).

15. (2 points) A spectrum from a Mars teaching dataset shows a mineral associated with past water alteration. What can and cannot be concluded?

answer: It can support a history of water-related processes under suitable geological context, but does not establish that life existed or that stable liquid water is present now. (2 points).

16. (2 points) Explain the difference between evidence for organic molecules on Bennu or comet 67P and proof of biology.

answer: Organic molecules can form through non-biological processes; their presence constrains chemistry and delivery pathways but is not by itself evidence of life. (2 points).

17. (2 points) Why can a planet in the TRAPPIST-1 system require atmospheric-retention evidence in addition to an estimated equilibrium temperature?

answer: An equilibrium temperature does not specify an atmosphere, greenhouse effects, stellar irradiation history or surface pressure; these affect stable liquids and habitability. (2 points).

18. (2 points) A planet is detected through periodic stellar radial-velocity shifts. What physical interaction produces the signal?

answer: The planet and star orbit their shared center of mass, creating a line-of-sight stellar velocity variation; other stellar variability must be considered. (2 points).

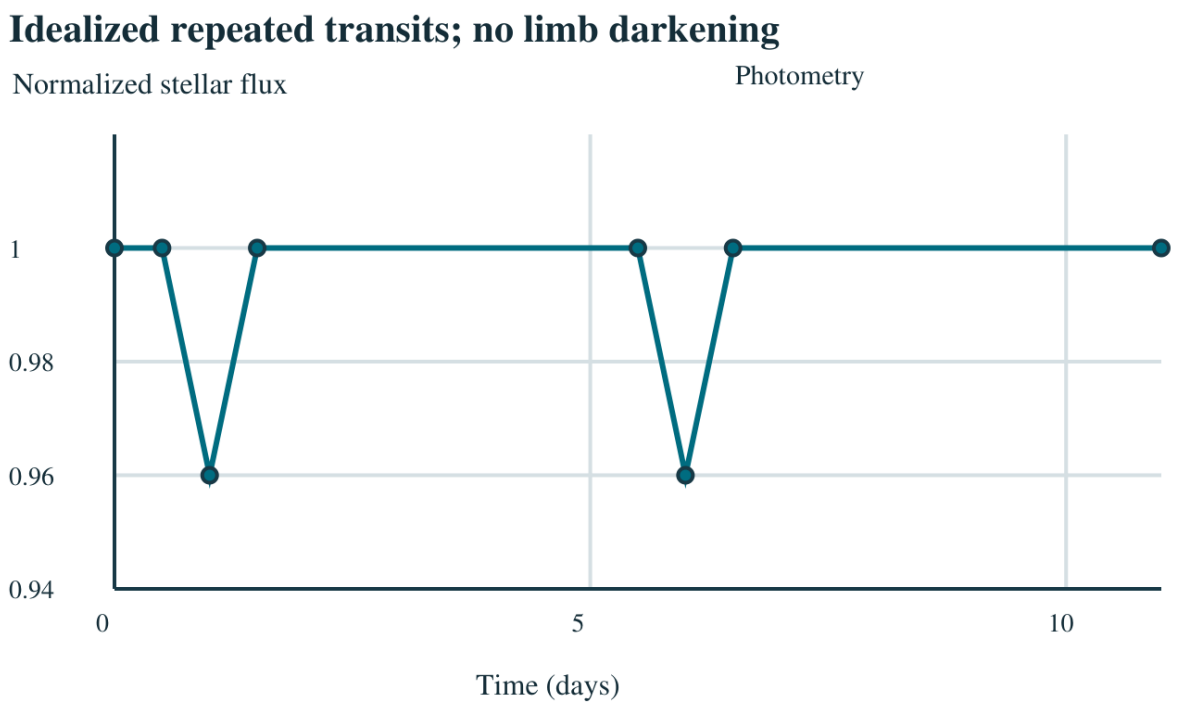
19. (2 points) Explain one engineering tradeoff between high spatial resolution and broad repeated mapping for an allowed mission.

answer: Finer sampling can require narrower swaths, greater data volume or observing time; a mission must balance resolution, coverage and revisit against its science goals. (2 points).

20. (2 points) Why is chemical disequilibrium potentially relevant to habitability but not a unique biosignature?

answer: A disequilibrium can provide usable free energy or motivate biological hypotheses, but geological, photochemical and transport processes can also maintain it. (2 points).

21. (2 points) Use Figure F1. Read the time between the two transit minima.



F1. Exoplanet transit light curve (original synthetic teaching figure)

answer: 5 days (2 points).

Numerical target 5; absolute tolerance 0.015. Accepted units: , days.

22. (2 points) Use Figure F1. Find transit depth as a percentage of baseline light.

Refer to F1. Exoplanet transit light curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: 4 % (2 points).

Numerical target 4; absolute tolerance 0.012. Accepted units: , %.

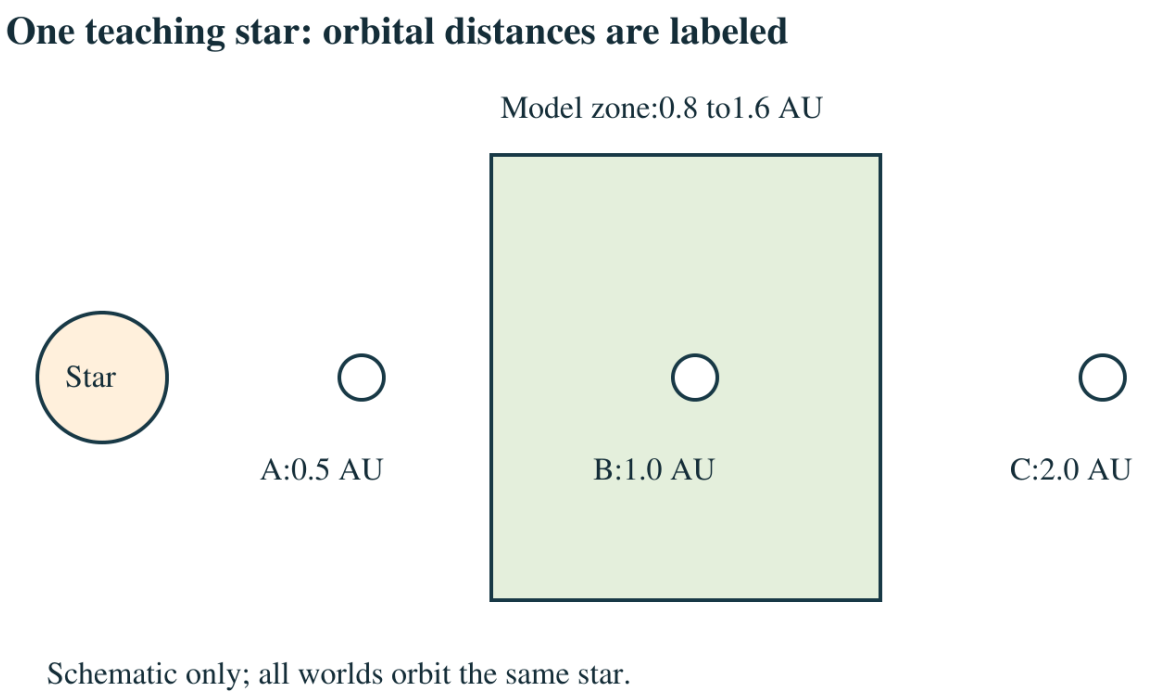
23. (2 points) Use Figure F1. Use depth=(planet radius/star radius)^2. The stellar radius is 5 units; find planet radius. Hint:0.2 squared=0.04.

Refer to F1. Exoplanet transit light curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1 units (2 points).

Numerical target 1; absolute tolerance 0.003. Accepted units: , units.

24. (2 points) Use Figure F2. Which candidate lies inside the indicated model zone?



F2. Habitable-zone placement (original synthetic teaching figure)

answer: B. (2 points).

25. (2 points) Use Figure F2. Explain why that placement does not establish habitability by itself.

Refer to F2. Habitable-zone placement (original synthetic teaching figure) shown with the first question in this figure group.

answer: Atmosphere, pressure, water inventory, surface conditions and energy balance still matter; orbital distance alone is insufficient. (2 points).

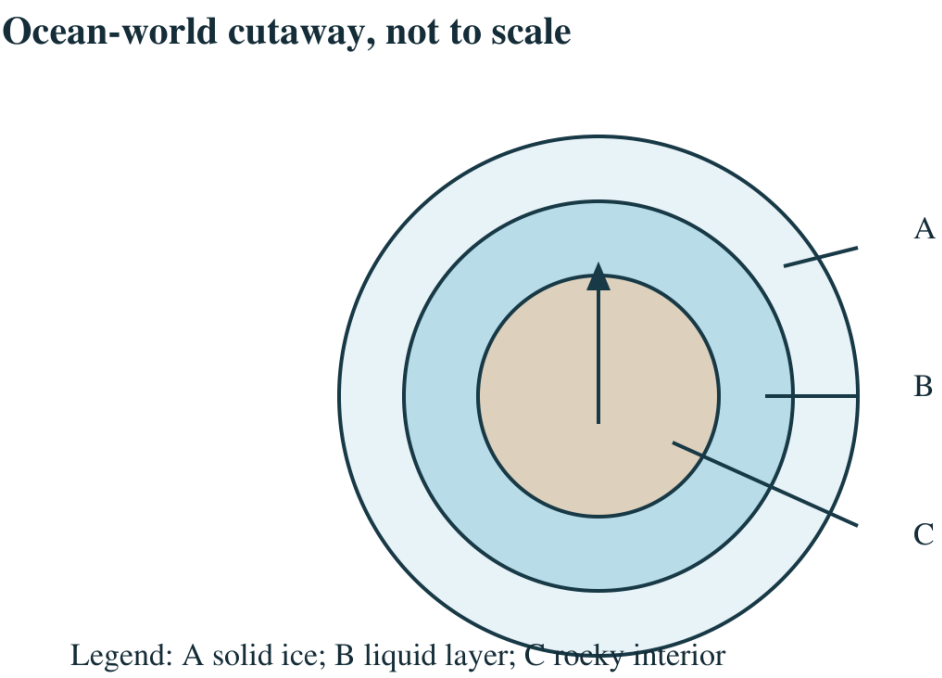
26. (2 points) Use Figure F2. Use the labeled orbital distances to find incident stellar flux at candidate B / flux at candidate C. Use inverse-square scaling.

Refer to F2. Habitable-zone placement (original synthetic teaching figure) shown with the first question in this figure group.

answer: 4 (2 points).

Numerical target 4; absolute tolerance 0.012. Accepted units: .

27. (2 points) Use Figure F3. Which labeled layer is the direct liquid-water habitat candidate?



F3. Icy-world interior hypothesis (original synthetic teaching figure)

answer: B. (2 points).

28. (2 points) Use Figure F3. Explain one interior energy source consistent with maintaining B beneath ice.

Refer to F3. Icy-world interior hypothesis (original synthetic teaching figure) shown with the first question in this figure group.

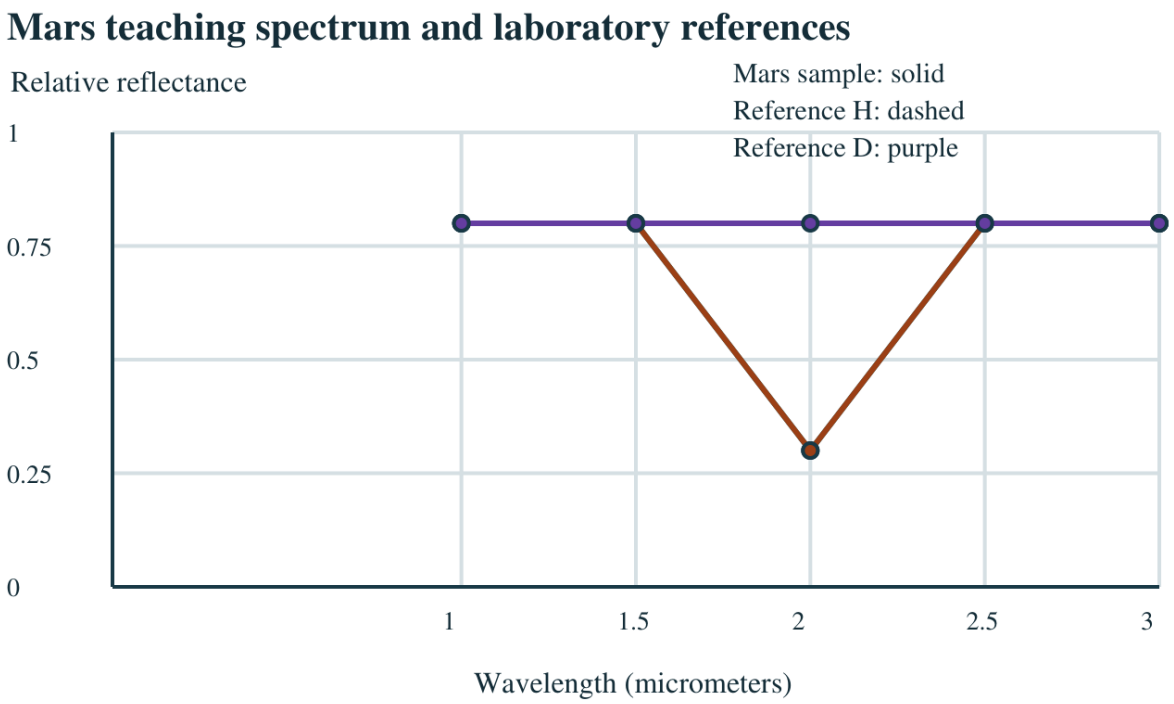
answer: Tidal dissipation or internal radiogenic heating could supply energy; sufficient heat and suitable pressure/composition must be established. (2 points).

29. (2 points) Use Figure F3. Would this interior hypothesis establish living organisms? Explain.

Refer to F3. Icy-world interior hypothesis (original synthetic teaching figure) shown with the first question in this figure group.

answer: No. Potential water and energy identify a candidate environment; direct or otherwise well-supported biological evidence would still be needed. (2 points).

30. (2 points) Use Figure F4. Which laboratory reference matches the plotted absorption?



F4. Matched mineral spectra (original synthetic teaching figure)

answer: H. (2 points).

31. (2 points) Use Figure F4. Read the sample reflectance at the absorption minimum.

Refer to F4. Matched mineral spectra (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.3 (2 points).

Numerical target 0.3; absolute tolerance 0.0009. Accepted units: .

32. (2 points) Use Figure F4. Explain why matching the hydrated-mineral reference supports water-related geological investigation without proving present life.

Refer to F4. Matched mineral spectra (original synthetic teaching figure) shown with the first question in this figure group.

answer: Hydrated-mineral evidence can support water-related alteration, but does not itself demonstrate a current liquid habitat or biology. (2 points).